Related work for each white-paper task

Draft, Oct 26 evening: Tasks 1.1, 2.1, 3 and Evaluation. Tasks 1.2 and 2.2 and the summary are still in progress, and citations have not yet had their independent check.

Prepared for Xinya Du, ONR FY27 YIP (Code 311)

Contents

Task 1.1: Deciding what to debate, counting shared mistakes once 1

Task 2.1: Debate before an impartial judge 4

Task 3: Rewards for contributions to trustworthy decisions 8

Evaluation: testbeds, red team and baselines 12

Task 1.1: Deciding what to debate, counting shared mistakes once

For Task 1.1, which weighs answers by track record and sends weakly supported questions to debate, the 20-page draft's cap on a correlated group's evidence, Proposition P1, holds and is tight. Its yes/no bound, $\log((1 - \lambda)/\lambda)$ for a shared hidden answer wrong with probability λ, is the classical optimal vote weight of one voter with accuracy $1 - \lambda$ (Nitzan and Paroush 1982; Shapley and Grofman 1984). P1 thus makes exact, for any group size, the classical advice to count a correlated bloc as one voter. **The most important change is to restate P1 in its general form, with the equal-likelihood formula as a corollary, and P2 as identification up to relabeling, anchored by planted cases.**

Prior work and its limits

Rater models assume independence.

- Dawid-Skene gives each rater a confusion matrix, the probability of each reported answer given each true answer, fitted without labels by EM, an iterative method; raters are independent given the truth (Dawid and Skene 1979).
- Zhang et al. 2016 (JMLR) start EM from a spectral estimate, a decomposition of three groups' joint answer statistics, with an error rate optimal up to a log factor; the proofs use independence and do not analyze its failure.
- Under dependence, such estimates "may be highly inaccurate" (Jaffe et al. 2016). Dawid-Skene rated GPT-5.2's valence labels 90.7% accurate, against 71.0% on a human gold set (Said and Villaneau 2026, LREC 2026 workshop), and trailed majority vote on MNLI, 70.7% against 72.0% (Kohli 2026, preprint).
- For Task 1.1: Zhang et al. supply the estimator, and these failures show the cost of assuming independence.

Copying and latent groups.

- Dong et al. 2009 (VLDB) detect copying among data sources from shared false values, since "sharing the same false value is typically a rare event", but warn that "sharing common data does not in itself imply copying".
- Jaffe et al. 2016 (AISTATS) place a hidden binary variable between the truth and each group of classifiers, the binary form of Task 1.1's model, and call it "in general not identifiable": different parameter settings can fit the same data.
- For Task 1.1: Jaffe et al. is the model's binary ancestor, and Dong et al. justify its "same unusual answers" signal.

LLM errors are shared.

- Larger, more accurate models have highly correlated errors, even across providers (Kim et al. 2025, ICML).
- Nine judges were worth 2.18 independent votes on MNLI, and 9.1% of the 319 items all nine agreed on were wrong, against about 0.02% under independence (Kohli 2026). Established aggregation closed at most 11% of the gap to the

accuracy expected from independent judges, even with correct answers; the datasets "remain classification or binary preference tasks".
- For Task 1.1: these measurements need labels, and the 11% suggests the lost accuracy is missing information, which motivates label-free estimation and debate for new evidence.

Selective discussion in LLM work.

- DOWN (Eo et al. 2025, preprint) starts a discussion only when one agent's confidence, its average token probability, is at or below a threshold; agreement plays no part. On Llama-3.3-70B it lifted the discussion method's score from 80.35 to 83.41, skipping 68.56% of discussions.
- iMAD (Fan et al. 2025, preprint) and SELENE (Verma et al. 2026, EACL 2026 Industry Track) trigger on hesitation cues, confidence-likelihood misalignment or semantic disagreement.
- Outside LLMs, disagreement-based monitoring missed 90.6% of dangerous under-predictions (McDonnell et al. 2026, preprint).
- For Task 1.1: the search found no LLM method that sends a unanimous answer to debate because the agreeing agents are dependent; its 2026 sweep was partial.

Why this method: the math behind it

Result	What it gives this task	Assumptions, and where it fails	Source
Optimal weighted vote: weight log(p/(1 - p)) for accuracy p	P1's yes/no cap, with p = 1 - λ	Independent voters, known accuracies, two options, equal priors; under dependence other rules can win	Nitzan and Paroush 1982; Shapley and Grofman 1984, Theorem II
Bloc rule: count each correlated bloc as one voter	P1 makes it exact for any bloc size	Called "a good approximation" when correlations between blocs are small	Shapley and Grofman 1984
Multi-class optimal weight log[(K - 1)a/(1 - a)] for accuracy a	P1's equal-likelihood corollary, a = 1 - λ	Errors independent given the truth; wrong answers equally likely	Ai et al. 2026, ICML, Theorem 3.1
Kruskal uniqueness for hidden-class models	P2	Three groups independent given the truth; a rank condition; positive priors; known groups; up to relabeling	Kruskal 1977; Allman et al. 2009, Annals of Statistics
Reject rule; Bayes-optimal deferral to an expert	Eq. 3, τ = 1 - ε_D - c/L	Calibrated posterior; fixed cost and expert accuracy; expert errors independent of the agents'	Chow 1970; Mozannar and Sontag 2020, ICML

P1 needs only that members depend on the truth through their group's shared hidden answer h, in any way among themselves; their answers X_g then favor y over y' by at most the bound below (researcher's derivation, reached to four decimals in simulation). The draft's multi-class formula, log[(|Y| - 1)(1 - λ)/λ], also assumes equally likely wrong answers: with three answers, λ = 0.3 and the shared mistake on one wrong answer 29 times in 30, a large, accurate group shifts the log-odds (log ratio of two answers' probabilities) by 4.25, beyond the formula's 1.54 but within the general bound, log(0.7/0.01) = 4.25 (researcher's computation). So state the general bound as P1, with the equal-likelihood formula, the optimal weight of Ai et al. 2026, as a corollary:

$$\log \frac{P(X_g \mid y)}{P(X_g \mid y')} \leq \max_h \log \frac{P(h \mid y)}{P(h \mid y')}$$

P2's condition is sufficient by Kruskal's uniqueness theorem as Allman et al. 2009 apply it (Corollary 2): the joint answers of three such groups pin down each group's answer distributions. It needs four additions:

1. Identification is only up to relabeling the answers, so Task 2.2's planted cases should anchor which answer is which.
2. Groups must be known or recoverable. Exact recovery by best fit is NP-hard, meaning no efficient algorithm is known (Jaffe et al. 2016, Lemma 3), so prefer groups observed through a shared base model or cited source.

3. What is identified is each group's answer distribution given the truth, which is all the posterior (each answer's probability given all answers) needs.
4. The condition is stronger than needed: two groups with linearly independent answer distributions (none a weighted sum of the others) plus a third whose distributions differ for every pair of truths suffice.

Eq. 3 follows from comparing expected losses: deciding now loses $L(1 - p)$ and debating loses $L\,\varepsilon_D + c$, so debate is better exactly when $p < \tau$ (p: the leading answer's posterior; L: the loss from a wrong decision; ε_D, c: the debate's error rate and cost). This is Chow's reject rule and the Bayes-optimal (lowest expected loss) rule for deferring to an expert, the debate, plus a cost (Chow 1970; Mozannar and Sontag 2020, Proposition 2); it assumes a calibrated posterior (probabilities that match observed frequencies) and debate errors independent of the agents' shared mistakes, which is why Task 1.2's judge shares no model with either side. With $\lambda = 0.3$ per group and a uniform prior, unanimous agents in k groups that err independently reach a posterior of at most 0.700, 0.845, 0.927 and 0.967 for $k = 1$ to 4, so at $\tau = 0.85$ ($\varepsilon_D = 0.10$, $c/L = 0.05$) unanimity within one or two groups goes to debate (researcher's computation).

Closest work and how this task differs

Work	What it does	What it assumes	What this task adds
Hossain et al. 2026 (preprint)	Measures judges' error dependence; a frontier trio is worth 1.41 judges	25-100 trusted labels; estimates "should not be assumed to transfer"	Label-free posterior; per-question choice to debate
CARE, Zhao et al. 2026, ICML	Separates quality from shared confounders, such as style, without labels; up to 26.8% lower error than majority vote	Quality is the dominant shared factor; judges not strategic	Shared mistakes as a hidden answer that can replace the truth; planted anchors; debate decision
OW-L and ISP, Ai et al. 2026, ICML	Label-free weights from answer co-occurrence	Errors independent given the truth; symmetric wrong answers	Group dependence; P1; debate decision
Balasubramanian et al. 2026 (preprint)	Label-free Ising (pairwise-interaction) and latent-factor models; proves independence-based aggregation keeps excess error as judges are added	Binary; item-level factor independent of the truth; identifiability not stated	Multi-class group answers; P1; P2; debate decision
DOWN, Eo et al. 2025	Discussion only when one agent's confidence is low	No agreement or dependence signal	Dependence-aware trigger that can send unanimous answers to debate
Jaffe et al. 2016, AISTATS	Hidden group variables between truth and classifiers; spectral group detection	Binary; "typically only an approximation"; exact group recovery NP-hard	Multi-class; P1; P2 with observed groups; audit flags; debate decision

The closest 2026 paper is CARE, which "works well when the dominant shared variation reflects quality", whereas in Task 1.1 a copied outdated source can dominate. The safest true claim is the combination: a shared hidden answer per group, a size-free cap matching classical vote weights, identifiability with observed groups and planted anchors, and a Bayes-optimal rule that can send unanimous answers to debate. The web-search budget ran out before the August-October 2026 sweep finished, so the novelty sweep should be re-run before submission.

What to borrow

Technique	Source	How to adapt it here
Three-group spectral start, then EM	Zhang et al. 2016	Use model families as groups; fit eq. 2 from the spectral start; cite the rate
Anchors to fix which factor is which	CARE, Zhao et al. 2026, Section 3.3	Use Task 2.2's planted cases as anchors
Shared false values as evidence of copying	Dong et al. 2009	Group agents by agreement on answers the posterior deems wrong

Technique	Source	How to adapt it here
Counting evidence sources	Zanolini 2026 (preprint)	Observe groups from cited documents and tool calls, making P2's known groups realistic
Effective number of judges (independent judges a panel is worth)	Kohli 2026; Hossain et al. 2026	Report it per testbed beside the posterior's implied number of independent votes
All-fail rate (share of items every model misses)	J. Chen 2026 (preprint)	Add a hard-item class every group can fail, since P2 assumes no item-level common cause
Learning when to defer to an expert	Mozannar and Sontag 2020	Learn ε_D per topic from logged debates and audits

Fixes to the white paper

- **Section 3, first sentence.** It cites Jaffe et al. 2016 for a limitation that paper partly removed, and "never change their minds" fails for WISE (Cherian et al. 2025, preprint, venue unconfirmed), which runs a modified Dawid-Skene across discussion rounds. Suggested: "Statistical methods estimate each rater's accuracy without an answer key (Dawid and Skene, 1979; Whitehill et al., 2009), and some allow dependence: sources that copy one another (Dong et al., 2009), groups of related classifiers (Jaffe et al., 2016), or shared biases among AI judges (Zhao et al., 2026). But each fixes the form of dependence in advance, treats answers as fixed reports from raters that do not act strategically, and none decides when the evidence is too weak and a question needs further scrutiny."
- **Task 1.1 text.** Cite Jaffe et al. 2016 and Dong et al. 2009. Suggested: "The core method extends the rater model of Dawid and Skene (1979) with shared error factors, as Jaffe et al. (2016) did for binary classifiers: agents that keep giving the same unusual answers share a factor (Dong et al., 2009), and their agreement adds little evidence."
- **Simulation (20-page draft).** It reports up to 0.85 for five independent agents at 70% plus k copies of a source wrong 30% of the time. The researcher computed the Bayes ceiling (the best accuracy achievable, counting the copy group once) as 0.837 for yes/no questions, and 0.914 or 0.937 with three or four options and wrong answers spread evenly. If questions are yes/no, 0.85 exceeds the ceiling: state the number of options or recheck the figure, and plot the ceiling with confidence intervals.
- **Citations (20-page draft).** CARE is Zhao et al. 2026, separate from Balasubramanian et al. 2026; DOWN is Eo et al. 2025. Add that CARE assumes quality is the dominant shared factor, and cite Chow 1970 and Mozannar and Sontag 2020 at eq. 3.

Draft related-work paragraph

> Rater models estimate each source's accuracy without an answer key (Dawid and Skene, 1979). Later work allows dependence: copying among sources (Dong et al., 2009), groups of related classifiers (Jaffe et al., 2016), and dependence among AI judges (Zhao et al., 2026; Balasubramanian et al., 2026). We extend the Dawid-Skene model with a shared hidden answer per group of dependent agents, as Jaffe et al. (2016) did for binary classifiers. We show that such a group, whatever its size, adds no more evidence than its shared source would alone. This makes exact the classical advice to count a correlated bloc as one voter (Shapley and Grofman, 1984). Track records can be recovered from answers alone when three known groups err independently given the truth and each group's answers differ enough across true answers (Kruskal, 1977; Allman et al., 2009). Planted cases fix which answer is which. A question goes to debate when its leading answer's probability falls below a cost-based threshold, the optimal rule for deferring to an expert (Chow, 1970; Mozannar and Sontag, 2020). Earlier work starts a discussion when one agent's confidence is low (Eo et al., 2025); ours can send even a unanimous answer to debate.

Task 2.1: Debate before an impartial judge

Of the closest debate systems we found, none has its judge re-run the debaters' exact tool calls, as Task 2.1's debate harness does at the first disputed step. Refereed delegation of computation justifies both the stop and the re-run (Canetti et al. 2011, CCS; Canetti et al. 2013, Information and Computation). When two servers disagree, the client bisects (halves the disputed range each round) to the first differing step, recomputes only that step, and gets the right result in logarithmically many rounds if one server is honest. **The most important change is to recast the stopping rule as refereed delegation: both representatives commit their steps and tool logs on one shared outline of sub-questions, and the loop stops at the first sub-question on which they differ.**

Prior work and its limits

Discussion erases disagreement. In Shu 2026 (preprint), seven LLM judges rated 600 code-correctness candidates, revised after seeing one another's answers, then voted. Unanimity rose from 39.5% to 95.2% after one round, while accuracy moved by less than one point. The PI's PRD is also a discussion, with two reviewers "trying to reach a final agreed review" and no judge (Li et al. 2024). For Task 2.1: let the judge's choice decide, and have each side commit before seeing the other's argument.

Standard debate relies on quote matching. In Khan et al. 2024 (ICML), debate raised judge accuracy from 48% to 76% for non-expert models and from 60% to 88% for humans, with quotes matched against the source text; without such matching, a debater "could simply create an alternative narrative". Kenton et al. 2024 (NeurIPS) leave tool use and code execution to future work, and in Elasky et al. 2026 (preprint) judges' verification fell once a critic appeared (gpt-oss: from 67–82% to 16–31%). For Task 2.1: a harness that re-runs tool outputs extends quote matching and does not depend on a diligent judge.

Debate theory assumes an honest side. Irving et al. 2018 (preprint) prove that with optimal play, debate with a polynomial-time judge can decide any problem in PSPACE (solvable with polynomial memory), but call such arguments "analogies only". Barnes and Christiano 2020 (Alignment Forum) found obfuscated arguments, long arguments whose few flawed steps neither side can locate, for every physics question studied, and reported no fix. Locating the first wrong step is hard even for strong models (Zheng et al. 2024, ProcessBench, venue unconfirmed), but LLMs can correct an error given its location (Tyen et al. 2024, Findings of ACL). For Task 2.1: the opponent locates the disputed step, the judge rules on it, and the guarantee is conditional, correct unless a named assumption fails, as in a debate safety case (Buhl et al. 2025, preprint).

Re-running needs reproducible evidence.

- What breaks it: LLM outputs vary even at temperature 0, because results depend on batch size, which varies with server load. Batch-invariant routines cut Qwen3-235B's 1,000 runs from about 80 unique outputs to one, at about 20% slower matrix multiplication (He 2025, blog post). Floating-point order also differs across hardware, and live APIs and web results change.
- Remedies in use: fixed floating-point order, which lets Verde apply refereed delegation to ML programs (Arun et al. 2025, preprint); cached responses and simulated APIs (StableToolBench, Guo et al. 2024, preprint); proofs that data came from a given secure (TLS) website (DECO, Zhang et al. 2020, ACM CCS).
- For Task 2.1: the harness could declare four evidence classes (a suggested design). Deterministic tools are re-run exactly, and archived sources against the logged snapshot; live sources are logged with call, time, response and hash; LLM calls are pinned to a model version at temperature 0, else treated as live. A live re-run that differs shows that the source changed, so the judge rules as of the logged time or reports the step unsettled, as for the vessel example's port report.

Why this method: the math behind it

Result	What it gives this task	Assumptions, and where it fails	Source
Refereed delegation: correct output if one server is honest; logarithmically many rounds; one step recomputed	The stopping rule and the re-run, exact at that step	Deterministic steps, a common input, one honest server, a secure hash. Fails: arguments share no program; inference steps cannot be recomputed; shared mistakes in agreed steps go unexamined	Canetti et al. 2011; Canetti et al. 2013; mechanics read via Ye et al. 2022, Nehab et al. 2024
Fraud proofs: bisection to one machine instruction, then a one-step proof; TrueBit adds forced errors	A deployed precedent; forced errors keep verifiers alert, like Task 2.2's planted cases	Deterministic steps; at least one verifier; timely moves. Fails: nondeterministic tools; timeouts that block the honest side	Kalodner et al. 2018 (USENIX Security); Teutsch and Reitwießner 2019 (preprint)
Refereed games: one round captures exactly PSPACE, polynomially many rounds exactly EXP (exponential time)	The ceiling for competing provers and a fast referee	Unbounded provers, an honest one playing optimally. Fails: bounded debaters (obfuscated arguments)	Feige and Kilian 1997 (STOC), read via Kol and Raz 2014

Result	What it gives this task	Assumptions, and where it fails	Source
Doubly-efficient debate: O(log T) rounds for a T-step computation, O(1) judge queries (Thm 5.1); one-round commit-and-point (Thm 5.3)	The AI-debate counterpart: efficient honest play, a one-step ruling	A shared program M; provers that predict the judge accurately. Fails: obfuscated arguments; a fallible judge	Brown-Cohen et al. 2024 (ICML)
Grounded semantics: argument games are sound and complete for it	A meaning for the losing rule and "unsettled"	Attacks set by the judge's rulings. Fails: arguments that add up independent evidence	Dung 1995 (AIJ); Prakken 2006 (KER)

Lean on refereed delegation for P4, which holds as stated if every argument for a false answer contains a false step. We suggest a sharper form (our synthesis, not a published result). Let M be the event that the correct side disputes none of the wrong side's false steps, B the event that it argues a false step itself, and ε_i the judge's error on disputed step i; a wrong decision needs both the wrong side to escape (M or a misruling) and the correct side to be struck (B or a misruling), since either alone strikes neither side or both, which is "unsettled".

$$P(\text{wrong decision}) \le \min\left\{P(M) + \sum_{i \in \text{wrong side}} \varepsilon_i,\ P(B) + \sum_{j \in \text{correct side}} \varepsilon_j\right\} \le P(M) + P(B) + \sum_{\text{all disputed } k} \varepsilon_k$$

The right-hand side is P4's union bound. Refereed delegation makes all three terms zero, up to hash collisions: determinism and a common input give P(M) = 0, an honest server P(B) = 0, and exact recomputation ε = 0. Each assumption that fails for LLM agents adds its term, and one part of the project targets each: Tasks 1.1 and 1.2 target P(B) and shared mistakes, the outline and stopping rule P(M), re-running ε on tool-backed steps only, and Task 2.2 representatives who throw a debate. With a shared outline one step is disputed, so the bound becomes the chance that the correct side errs there plus the judge's error on it.

For P4's losing rule, lean on grounded semantics, with steps as arguments and disputes as attacks: a move is out if some reply to it is in, and in if all replies attacking it are out (Prakken 2006, after Dung 1995). A step the judge cannot rule on leaves both sides undecided, the standard meaning of "unsettled"; if both sides have a struck step, both are out, and the draft should name that case unsettled too, unless a burden of persuasion declared in advance sends ties against one side (Prakken and Sartor 2007, ICAIL). Because one struck step sinks an argument, the rule breaks for arguments that add up independent evidence, where prover-estimator debate's probabilities fit better (Brown-Cohen et al. 2025, preprint).

Closest work and how this task differs

Work	What it does	What it assumes	What this task adds
Jiang et al. 2026 (ICML)	Two consultants name a disputed premise or step and may switch, until they agree or hit a turn limit; a passive judge reads the transcript; 62.1% vs 49.2% for debate with weak judges	Consultants that "produce reliable reasoning"; at least one correct; no tools	Impartial, possibly dishonest representatives; a stop at the first disputed step; a judge who re-runs calls and decides without consensus
PROClaim (Chowdhury et al. 2026, preprint)	Courtroom roles; fixed PubMed index; an LLM arbiter admits evidence; 3-judge majority; 81.7% on Check-COVID vs 85.8% for one GPT-5-mini call with retrieval	Trusted arbiter and panel; judges do not re-retrieve; INCONCLUSIVE counts as SUPPORT	Exact re-running at the disputed step; an impartial judge; an "unsettled" output
DynaDebate (Li et al. 2026, preprint)	Step-by-step critique; code or search only "when a deadlock persists", as a reference opinion; answer by extraction or vote	Diverse paths reduce shared errors	The opponent names the disputed step; the re-run result is evidence for a judge whose choice decides

Work	What it does	What it assumes	What this task adds
Tool-MAD (Jeong et al. 2026, preprint)	Wikipedia retrieval vs live web search; stops on consensus, else a judge scores faithfulness and relevance without calling tools	Consensus means done; live search	The judge re-runs exact calls against logged snapshots
Prover-estimator debate (Brown-Cohen et al. 2025)	Theory: a prover splits a claim, an estimator gives probabilities, the judge rules on one leaf; an LLM version is left open	Stability, equilibrium, a correct leaf judge	An implemented harness with tools and re-running
Standard debate (Khan et al. 2024; Kenton et al. 2024)	Two debaters before a weaker judge; quotes or passages matched by string	Information asymmetry; honesty-tuned debaters; text only	Re-run tool outputs; shared mistakes; collusion audits

The closest 2026 paper, Jiang et al. 2026, uses no tools and leaves dishonest models untested; the safest true claim is that, in the work we found, no debate or discussion system has its judge re-run the debaters' exact tool calls. The search was not exhaustive, because the web-search budget ran out before the August–October 2026 sweep, so the novelty sweep should be re-run before submission.

What to borrow

Technique	Source	How to adapt it here
Commit, then find the first differing step	Canetti et al. 2013; Teutsch and Reitwießner 2019	Both sides commit steps and a hash-chained (tamper-evident) tool log; the harness finds the first differing sub-question, bisecting only inside long tool traces
Shared outline	Jiang et al. 2026 (collaborative phase)	A short collaborative phase or a testbed template fixes the sub-questions
Forced errors	Teutsch and Reitwießner 2019	Planted debates with forged tool outputs flag judges that skip re-running (Task 2.2)
Reproduced-or-not tags	Khan et al. 2024	Tag each tool output as reproduced or not; the judge trusts only the former
Burden of persuasion	Prakken and Sartor 2007; Chowdhury et al. 2026	Declare per question type which side loses a tie; otherwise "unsettled"
Step parsing of search calls	HiPRAG (Wu et al. 2026, ICLR)	Each parsed search call becomes a logged, re-runnable evidence unit

Fixes to the white paper

- **Shu 2026.** Shu's panels revise for 1–3 rounds, then vote; Shu does not say systems "discuss until they agree". Suggested: "In multi-agent discussion, agents read one another's answers and revise them; a single round can make a panel unanimous without making it more accurate (unanimity rose from 39.5% to 95.2% while accuracy moved by less than one point), erasing the disagreement that exposes errors (Shu 2026)."
- **Stopping-rule citation.** Irving et al. do not define "the first step on which they differ". Suggested: "the two answer the same sub-questions in order, and the loop stops at the first one on which they differ, as in refereed delegation of computation, where a client settles a dispute between two servers by recomputing one step (Canetti, Riva and Rothblum 2013; Brown-Cohen, Irving and Piliouras 2024)."
- **"Fixed rounds of text".** This overstates. Suggested: "Most LLM discussions run a set number of rounds, or stop once the agents agree, among a fixed set of models; some retrieve documents or run code, but their judges read the transcript and do not reproduce the tool calls (Jeong et al. 2026; Chowdhury et al. 2026)."
- **Agent-as-a-Judge.** Its best system has no execution module (Zhuge et al. 2025, ICML), so re-running is the new part. Suggested: "the judge, an agent like those that already evaluate other agents' work (Zhuge et al. 2025), re-runs the call in the sandbox."
- **Irving claim ("Why it will succeed").** Lead with refereed delegation (Canetti, Riva and Rothblum 2013) and fraud proofs (Kalodner et al. 2018), as in the draft paragraph below, then add "with optimal play" and the obfuscated-arguments caveat. Suggested: "In debate with optimal play, a judge who rules on one step can settle questions far

beyond its own ability (Irving et al. 2018; Brown-Cohen et al. 2024). These results assume an honest side that can find the flawed step, which can fail for long arguments (Barnes and Christiano 2020), so our judge may report a ruling as unsettled."

- **Kenton et al. 2026.** Its headline is that debate training reduced reward hacking; invented quotes appeared only when the authors weakened the judge (Kenton et al. 2026, preprint). Suggested: "even debate training, which reduced reward hacking overall, taught a critic to invent quotations when the judge was weakened (Kenton et al. 2026)."
- **Limitations.** Add, citing Hu et al. 2026 (ACL): "Re-running evidence does not help with true but selectively chosen facts (Hu et al. 2026) or with live sources that change between the debate and the ruling, so the harness logs each response with its time and source."
- **Bibliography.** Feige and Kilian 1997 and the CCS 2011 protocol details were read only through later papers; confirm them in the originals before citing details.

Draft related-work paragraph

> In multi-agent discussion, agents read one another's answers and revise them; a single round can make a panel unanimous without making it more accurate (Shu, 2026). Standard debate grounds claims in quotations matched against a source text (Khan et al., 2024; Kenton et al., 2024). Some systems let agents retrieve documents or run code, but in the work we found, no judge reproduces the debaters' tool calls (Jeong et al., 2026; Li et al., 2026; Chowdhury et al., 2026). Our stopping rule follows refereed delegation of computation. When two servers disagree, a client recomputes only the first step on which they differ, and the result is correct if one server is honest (Canetti et al., 2013). Blockchain fraud proofs use this method (Kalodner et al., 2018), and debate theory gives its counterpart for AI (Brown-Cohen et al., 2024). Our judge likewise re-runs tool calls at the disputed step. Our design, like debate theory, assumes an honest side that can find the flawed step. This can fail for long arguments (Barnes and Christiano, 2020), so our judge may report a ruling as unsettled.

Task 3: Rewards for contributions to trustworthy decisions

In Task 3, which trains the agents by reinforcement learning (RL) on the group's decisions, every label-free reward the search found, apart from chance-corrected agreement, has an optimum that does not depend on correctness. Rewarded for matching its own majority vote, a model gave one template answer to nearly every prompt after 1,200 steps (Shafayat et al. 2025, preprint). The classical basis for the method is peer prediction: scoring agreement with a reference answer minus the agreement expected by chance pays zero on average to answers that ignore the question, and cancels the reference's bias toward one answer (Dasgupta and Ghosh 2013; Liu and Guo 2020, ICML). **The most important change is to stop crediting both debaters for moving the judge toward the decision, which pays the side assigned to the other answer to argue badly; debaters should keep the reward for their own side, or earn only a bounded bonus for tool-backed messages the judge re-runs.**

Prior work and its limits

Label-free rewards collapse.

- Self-rewarded training (SRT) rewards answers that match the majority of the model's own 16-64 samples (Shafayat et al. 2025). On Qwen2.5-Math-7B it first neared RL on true answers (0.32 against 0.33, trained on MATH-12K); on DAPO it declined after about 400-600 steps and collapsed by 1,200.
- Under GRPO, an RL method that compares samples drawn for the same question, random rewards raised Qwen2.5-Math-7B on MATH-500 by 21.4 points, against 29.1 for true rewards (Shao et al. 2026, ICML). Such rewards often did not help Llama3 or OLMo2.
- For Task 3: the main experiment needs a random-reward control and a second model family.

Agreement among several models.

- Co-RL (Yang et al. 2026, preprint, August 2026) arranges models in a ring and rewards each for matching one peer's majority vote. It gains 3.0-8.6% on average over 7 text benchmarks, with no chance correction and no protection against coordinated answers.
- For Task 3: Co-RL is the closest baseline for the first-answer reward (eq. 4). Without its chance term and dependence weighting, eq. 4 is close to Co-RL with many peers, so Co-RL should be run as an ablation.

Judge rewards get hacked.

- Kenton et al. 2026 (preprint) train a generator and a critic by standard debate before a frozen, weaker judge on AIME-

like math. Peak accuracy was 0.7474, against 0.7263 for RLAIF (RL on an AI judge's approval) and 0.7730 for RLVR (RL on automatically graded answers).
- The RLAIF model learned to impersonate the judge, in about 90% of samples at 30% of training. Without word limits the critic exploited the judge's preference for long answers, and its fabricated quotes were reported only before a deliberately weakened judge.
- Training debaters by self-play DPO (preference training) on the judge's probabilities, with quote validation, raised judge accuracy about 4 points (Arnesen et al. 2024, preprint). A one-seed LessWrong post finds judge alignment falling sharply after about 50 steps (Lennie et al. 2026).
- For Task 3: the debate-win baseline needs soft judge rewards, quote validation, word limits and a frozen weaker judge.

The PI's process rewards.

- FG-PRM (Li et al. 2025, Findings of EMNLP) is six step-level classifiers, one per hallucination type, trained on synthetic errors written from gold solutions. The paper uses them to detect errors and rerank sampled solutions; the parts read show no RL training. Choosing among 64 samples, it scored 0.94 on GSM8K and 0.57 on MATH, against 0.91 and 0.54 for Math-Shepherd, while its average detection F1 (combining precision and recall) is below ChatGPT's: 0.499 against 0.513 on synthetic data, 0.484 against 0.531 on human-annotated data.
- HiPRAG (Wu et al. 2026, ICLR) adds a reward for each search decision (a process reward) to outcome and format rewards during RL. It averages 65.4% (3B) and 67.2% (7B) on seven QA benchmarks and cuts unnecessary searches from over 27% in earlier baselines to 2.3%.
- For Task 3: credit moves from one agent's steps to many agents' messages. The step score becomes the change in the judge's belief, and the group decision replaces gold solutions, which FG-PRM names as its limit.

Why this method: the math behind it

Result	What it gives this task	Assumptions, and where it fails	Source
Uniform error rate η among K answers scales the clean risk by 1 - Kη/(K-1) (Thm 1)	P6(i) for K answers	Symmetric errors; η < (K-1)/K	Ghosh et al. 2017, AAAI
Unequal error rates ρ0 (false yes) and ρ1 (false no) shift the optimal threshold (Lemma 7)	Plain agreement inherits the reference's bias	Errors depend only on the true answer; ρ0 + ρ1 < 1	Natarajan et al. 2013, NIPS
Peer loss subtracts the loss on a random pairing: noisy loss is 1 - ρ0 - ρ1 times clean loss (Lemma 2); weight α* restores the optimum (Thm 4)	Bias cancellation in P6(ii); α*	Yes/no; errors independent of the question given the truth	Liu and Guo 2020
Agreement minus an other-task statistic pays uninformed reports 0; a trusted agent makes truthful effort "essentially the only equilibrium"	Structure of eq. 4; trusted anchors	Tasks alike in advance; coordination on cheap signals still pays	Dasgupta and Ghosh 2013; Shnayder et al. 2016; Gao et al. 2016
Dynamic potential-based shaping preserves optimal policies and equilibria	Eq. 5 for the group	Target fixed in advance; potentials logged when visited; every agent shaped on every transition	Devlin and Kudenko 2012, AAMAS
A policy's advantage is an optimal process reward	Message credit as its advantage	Judge's probabilities forecast its own final view	Setlur et al. 2024; Jia et al. 2025, preprint

For P6, let the reference, the answer the Task 1.1 posterior favors without the agent's own group, err at rate η among K answers, symmetrically and independently of the agent given the true answer Y. If the chance term scores the same answer a against the reference on a random hidden question from the same topic T, eq. 4's expected reward, ignoring confidence weights, is (researcher's derivation):

$$\mathbb{E}[\, r \mid a, q \,] = \left(1 - \frac{K\eta}{K-1}\right)\Big[P(Y = a \mid q) - P_T(Y = a)\Big]$$

- **P6(i) holds in a special case.** Dropping the base-rate term leaves plain agreement, whose factor is 1 - 2η only for yes/no answers with symmetric errors, a reference independent of the agent given the truth, and η < 1/2. With K answers it is 1 - Kη/(K-1) (Ghosh et al. 2017); with asymmetric errors, plain agreement inherits the reference's bias (Natarajan et al. 2013).
- **P6(ii) holds for answers that ignore the question.** The chance term pays them zero on average, including a fixed answer colluders agree on (Dasgupta and Ghosh 2013; Shnayder et al. 2016), and cancels the reference's bias, leaving the factor 1 - ρ0 - ρ1 (Liu and Guo 2020). Under symmetric errors plain agreement is already accuracy-optimal, so the chance term buys protection against collapse and collusion.

Four problems remain:

- **Base rates.** At full weight the best answer is the one the evidence lifts most above its topic base rate, which is the most likely answer only on balanced topics. Balance batches on the posterior answer within each topic, or use Liu and Guo's weight α*, computed from the error rates and the answer imbalance.
- **Crediting.** As worded, the term "does not depend on the current answer", so under GRPO it cancels exactly when shared across a question's samples. It must be credited to the answer it scores.
- **Cheap signals.** Colluders who coordinate on cheap features of a question, such as wording or length, still earn reward; Gao, Wright and Leyton-Brown 2016 show this is unavoidable when such signals exist.
- **All-ignore outcome.** If every agent ignores the question, all rewards are zero and learning stops, so trusted anchors are needed: tool-backed decisions and human audits (Dasgupta and Ghosh 2013, contribution iii).

For eq. 5, let Φ(h) be the judge's probability of the target decision after transcript h, logged as each message is read. Each message earns the change it causes, so over n messages the credits telescope:

$$\sum_{t=1}^{n} \left[\Phi(h_t) - \Phi(h_{t-1})\right] = \Phi(h_n) - \Phi(h_0)$$

For the group this is potential-based shaping (adding the change in a state's score to the reward, which leaves the best policy unchanged) of one end-of-debate reward, if the target is fixed before this debate's ruling (Devlin and Kudenko 2012). Each agent gets only its own credits, the group's gain minus the others', which favors the last speaker, so per-agent invariance fails and the draft is right that policy invariance does not carry over. If the judge's probabilities are calibrated forecasts of its own final view, each credit is that message's advantage, its effect on the expected final probability, the process reward that Setlur et al. 2024 and Jia, Rakhlin and Xie 2025 justify.

Closest work and how this task differs

Work	What it does	What it assumes	What this task adds
Co-RL, Yang et al. 2026	Each model in a ring matches one peer's majority vote	Model diversity prevents shared errors	Left-out, dependence-weighted reference; chance term; message credit; P6
Kenton et al. 2026	Generator-critic debate RL against RLAIF, frozen weaker judge	Word limits balance the game	Target is not this judge's ruling; first answers rewarded without debate; audits
Qiu et al. 2026	Peer-prediction log-score with a subtracted marginal; DPO training	Large, diverse pools; collusion not considered	Agreement with a decision; audits and dependence weighting
SRT and TTRL, preprints, 2025	One model matches its own majority	Prior knowledge; early stopping	Reference from other groups; zero pay for constant answers
ACC-Collab, Estornell et al. 2025, ICLR	Turn credit: chance that a discussion reaches the correct answer	Ground-truth labels	The same credit form without labels

The closest 2026 paper is Co-RL, so the safest true claim is the combination: a chance-corrected reward against a left-out, dependence-weighted reference with stated guarantees (P6), plus credit for each message toward a decision that needs no labels, which the search did not find in prior work. The web-search budget ran out before the August-October 2026 sweep, so the novelty sweep should be re-run before submission.

What to borrow

Technique	Source	How to adapt it here
α^*-weighted chance term	Liu and Guo 2020, Thm 4	Estimate $\rho 0$ and $\rho 1$ from Task 1.1's confusion matrices, or balance topics
Spot-checks against ground truth	Gao et al. 2016	Use human-audited real decisions as an occasional training reward
Potentials logged at visit time	Devlin and Kudenko 2012	Store the judge's probability as each message is read; never recompute with a later judge
Counterfactual credit	Foerster et al. 2017, COMA	Re-query the judge without the message, or with a resampled one
Collapse controls	Shafayat et al. 2025; Zhang et al. 2026, Co-rewarding, ICLR	Early stopping on audited decisions; slowly updated reference agents

Fixes to the white paper

- **Kenton et al., Section 3.** Misleading as written. The paper's main finding, in its title, is that "Debate Training Reduces Reward Hacking in RLAIF" (peak accuracy 0.7474 against 0.7263). Fabricated quotes came from the debate critic and were reported only before a deliberately weakened judge (chain of thought off, fewer than 50 visible words); the approval-trained model learned to impersonate the judge (Kenton et al. 2026).
- **Shafayat et al. and Wen et al., Section 3.** Both are fair but should be more precise. Shafayat's majority is the model's own samples; accuracy first neared ground-truth training (0.32 against 0.33), then collapsed to one template answer, and a constant answer is optimal for that reward. Wen et al. concern human approval and time-limited evaluators (3-10 minutes), whose false positives rose by 24.1% on QuALITY and 18.3% on APPS (Wen et al. 2025; confirm the ICLR 2025 venue).
- **Suggested Section 3 wording.** "Nor can agents yet keep improving without an answer key. Rewarding agreement with a model's own majority vote helped at first, then collapsed accuracy as the model learned to give one answer to every question (Shafayat et al., 2025). Rewarding an AI judge's approval taught a model to impersonate the judge, and even training by standard debate, which curbed this, let a critic fabricate quotes before a weakened judge (Kenton et al., 2026). Rewarding human approval taught models to convince time-limited evaluators that wrong answers were right (Wen et al., 2025). Newer methods that reward agreement among several models report stable training through model diversity, but offer no guarantee against shared or coordinated answers (Yang et al., 2026)."
- **P6.** Restate it with the assumptions and the four problems above. Plain wording for the panel: "If the reference answer is wrong a fraction η of the time, independently of the agent, rewarding agreement with it trains the agent in the same direction as the true answer, at strength $1 - 2\eta$ for yes/no questions. Subtracting how often the same answer matches the reference on an unrelated question from the same topic pays, on average, nothing to any answer that ignores the question, including a fixed answer that colluders agree on, and it removes the reference's preference for one answer."
- **Eq. 5 and the Task 3 text.** "Credits each message for helping reach the decision rather than for winning" holds only if the target comes from outside this debate's ruling, such as the Task 1.1 posterior plus re-run tool evidence; otherwise the credit is a win-margin reward. Crediting both debaters toward the decision pays the one assigned to the other answer to argue badly, the collusion Task 2.2 flags, and pays the honest side to concede when the decision is wrong. Fix: debaters keep the standard zero-sum reward, the judge's final probability of their assigned side (Arnesen et al. 2024), and the decision reward trains the judge and first answers; or debaters earn only a bounded bonus for tool-backed messages the judge re-runs. Compare both in the main experiment.
- **Main experiment.** Suggested: "We compare this reward with majority agreement, Co-RL (Yang et al., 2026), peer prediction (Qiu et al., 2026), rewards for winning standard debates (Arnesen et al., 2024), random rewards (Shao et al., 2026) and RL on true answers, in two model families."
- **Citations.** Cite Co-RL as Yang et al. 2026, arXiv 2608.17253, and Lennie et al. 2026 as a LessWrong post by "lennie, joanv, Shi and J. Pfau" (MATS 9.0, July 2, 2026). Casheekar and Tangirala 2026 (preprint) trains PPO agents and is not a study of LLMs. Describe FG-PRM as error detection and reranking, and do not claim its detection F1 beats ChatGPT's.

Draft related-work paragraph

> Rewarding agreement with a model's own majority vote helped at first, until the model gave one answer to every question (Shafayat et al., 2025). Rewarding agreement with a peer model's majority vote works without labels (Yang et al., 2026), but it also pays for a fixed answer that all the models give. Rewarding an AI judge's approval taught a model to impersonate the judge. Standard debate curbed this, yet a critic still fabricated quotes before a weakened judge (Kenton et al., 2026). We reward each first answer for matching a reference built without the agent's own group, minus how often the same answer matches the reference on another question from the same topic (Dasgupta and Ghosh, 2013; Liu and Guo, 2020). Answers that ignore the question then earn nothing on average. Prior turn-level credit for discussion needs the correct answer (Estornell et al., 2025). We credit messages by how far they move the judge, as process rewards credit progress (Setlur et al., 2024). Debaters are credited only toward their own side or for tool evidence the judge re-runs, so neither is paid to argue badly. This extends our step-level rewards (Li et al., 2025; Wu et al., 2026) to many agents' messages.

Evaluation: testbeds, red team and baselines

Each testbed can start from public data with known answers; the maritime scenarios and planted debates (debates on known-answer questions) must be built. MEQA as published builds each question from one document (Li et al. 2024), so multi-document maritime questions extend it. **The most important change is a red team that chooses when to attack, at four levels of attacker power (ACL-1 to ACL-4 in Korbak et al. 2025, preprint).** Such attack selection cut safety, the chance that no attack gets through, from 99% to 59% at a 0.5% audit budget (Schaeffer et al. 2026, preprint).

Testbed A: math and code with a shared flawed hint

Testbed A borrows a protocol that poses each question with and without a hint and keeps pairs that switch to the hinted answer (Chen et al. 2025, preprint). Giving one flawed hint to k of N agents makes the group sharing a mistake known by construction, and each agent's switch rate shows how easily it is misled.

When a hint changed an answer, the reasoning mentioned it in only 25% (Claude 3.7 Sonnet) and 39% (DeepSeek R1) of cases on average (Chen et al.). Shared errors must therefore be inferred from who agrees with whom, as in Task 1.1.

Dataset or protocol	What it gives	Size	Access	How we would use it
Paired hints (Chen et al. 2025; Turpin et al. 2023)	Six hint types (Chen); accuracy drops of up to 36% (Turpin)	MMLU, GPQA; 13 BIG-Bench Hard tasks	Preprints; public data	One hint for k of N agents; vary hint type and k
ProcessBench (Zheng et al. 2024)	Earliest wrong step, marked by doctoral-level annotators	3,400 solutions	Preprint; HuggingFace; no license stated	Disputed-step labels (P4); 18.8–51.8% of correct-answer solutions in harder subsets hide a wrong step
CodeCrash; ImpossibleBench	Misleading hints and structural changes (23.2% average drop); specifications contradicting unit tests	1,279 questions; LiveCodeBench, SWE-bench tasks	NeurIPS 2025; ICLR 2026	Shared hint for code; cheating incentive, unit tests as automatic audits
Password-locked models; sandbagging	Skill hidden unless a password appears; prompted underperformance	n/a	Preprints	Agents of set skill, re-measured on held-out items
MATH; GSM8K; LiveCodeBench	Problems with answers or unit tests	12,500; 8.5K; 400	Public	Source items; LiveCodeBench's date split limits contamination

Testbed B: multi-document questions and maritime scenarios

Testbed B asks questions that need several documents, such as whether a vessel has left port, and plants outdated or forged reports among them. Its closest competitor is MADAM-RAG, in which agents discuss conflicting documents and an aggregator combines their answers (Wang et al. 2025, venue not confirmed).

AIS, the position and status reports that ships broadcast, may be spoofed (MARAD advisory 2026-015), and 72–76% of industrial fishing vessels are not publicly tracked (Paolo et al. 2024, Nature). Real AIS tracks should therefore only seed scenarios, and the scenario generator must define each true answer.

Dataset or protocol	What it gives	Size	Access	How we would use it
MEQA	Multi-hop event questions (chaining facts) with explanations, one document each	2,243 questions, 211 documents	GitHub, MIT license	Base pipeline, extended to several documents
RAMDocs, MADAM-RAG	Ambiguous queries, misinformation, noisy documents	Not in abstract	Preprint	Required baseline; gains up to 15.80% (absolute) on FaithEval
ConflictBank; ConfRAG	Misinformation, temporal and semantic conflicts; real contradictions in web text	553,117 question-answer pairs; 1,814 questions	Preprint; ACL 2026	Outdated-schedule templates; held-out real conflicts
MarineCadastre; Danish AIS	Positions, speeds and status codes	US waters from 2009; Danish waters	CC0 (public domain) per repository; Danish terms not stated	Seed tracks; departure as status 5 (moored) to 0 (under way), after confirming code ranges
Global Fishing Watch	Port visits: entry within 3 km of an anchorage, exit beyond 4 km	Global	API token; non-commercial only	"Left port" labels; low-confidence visits may be false
Paolo et al. 2024; xView3-SAR	Satellite vessel detections matched to AIS	2017–2021; nearly 1,000 radar scenes	"Freely available"; registration; no license seen	Public stand-in for the satellite-image audit

Testbed C: scientific claims and peer review

Testbed C has agents judge scientific claims and papers, with replication outcomes as human audits. The best human forecasts of these outcomes were right 71–78% of the time (Dreber et al. 2015; Gordon et al. 2021; SCORE), while SCORE's machine forecasts "were not consistently effective".

Older outcomes, such as the 2015 Open Science Collaboration replications (summary), may be in training data, so 2026 releases with anonymized claims make safer audits. The labels are also noisy: in Many Labs 2, effects varied significantly across samples for 11 of 28 findings (Klein et al. 2018).

Dataset or protocol	What it gives	Size	Access	How we would use it
SCORE; Institute for Replication (both 2026)	Replication, reproduction and reanalysis outcomes; SCORE adds forecasts	49% of 164 papers replicated; 110 papers	OSF, Zenodo; formats and licenses not checked	Fresh human-audit labels
Gordon et al. 2021	Replication outcomes with prediction-market (betting) and survey forecasts	103 replications	PLOS ONE; R package pooledmaRket	Human collective baseline; markets 73% accurate
Schroter et al. 2008; SPECS (AAAI-26 pilot, preprint)	Major errors planted in papers; LLM-perturbed papers with a detection score	9 errors in each of 3 papers; 783 papers	JRSM; SPECS release not confirmed	Planted cases; reviewers found 2.58 of 9 errors, AI detection rose from 0.43 to 0.64
SPOT	Real errors that led to errata or retractions	83 papers, 91 errors	Preprint	Held-out real errors; the best model found at most 21.1%
PeerRead; NLPeer	Reviews; PeerRead adds accept or reject decisions	14.7K drafts, 10.7K reviews; over 5,000 papers, 11,000 reviews	NAACL 2018; preprint; licenses not checked	Disagreement among human reviewers

Dataset or protocol	What it gives	Size	Access	How we would use it
Agents4Science	Reviews by three LLMs per paper; human scores for 79 papers	253 reviewed papers	Preprint; reviews public	Correlated AI reviewers (Pearson 0.48)

Red team

AI control evaluations let a red team attack a fixed protocol with a small audit budget, 2% of solutions in Greenblatt et al. 2024 (ICML). Of the five levels in Korbak et al. 2025, ACL-5 is out of scope: its authors say "we cannot determine rules for a control evaluation".

The red team must include attack selection, recognizing planted debates, sandbagging (arguing badly on purpose), prompt injection (instructions hidden in text the judge reads), true but selectively chosen facts, steganography (hidden signals between colluders) and attacks spread across agents. In the table, only Hu et al. (ACL 2026) has a confirmed venue.

Source	What to borrow	How it maps to levels 1–4
Korbak et al. 2025	Red-team powers scaled to model capability	1: prompts, poisoned documents; 2: fine-tuned covert encoding; 3: goals held in weights, copies coordinating; 4: any trained attack
Schaeffer et al. 2026	Attack selection as classification; safety hinges more on the monitor's false-positive rate than its true-positive rate	1–4: cheat only when no audit is expected; their attacker was only prompted
Terekhov et al. 2025; Hu et al. 2026	Prompt injection hidden in outputs; collusion with true evidence fragments (70.6–74.4% attack success)	1–2: attacks on the judge that re-running evidence does not catch
Taylor et al. 2025; van der Weij et al. 2024	Selective sandbagging; a blue team could not reliably spot it (Taylor)	2–4: debaters or judges that argue badly on purpose
Needham et al. 2025; Hubinger et al. 2024	Telling evaluations from real use (AUC 0.83 for Gemini-2.5-Pro, 0.92 for humans; 0.5 is chance); backdoors that survive safety training	2–3: behave well whenever a debate looks planted; report planted-versus-real AUC
Kitaoka et al. 2026; Pang et al. 2026	Attacks spread across agents (7% flagged, against 87% for one agent); covert channels needing no shared key	4: attacks spread across debates; audits must look beyond transcripts

Baselines and metrics

Majority vote at equal compute is the bar to beat, since voting explains most gains credited to discussion (Choi et al. 2025, NeurIPS). For categorical answers, averaging with equal weights is essentially majority vote; keep it only for numeric scores, and add the rater model and multi-family panel below. For free-form answers, add universal self-consistency, where an LLM picks the most consistent sampled answer (Chen et al. 2023, preprint).

Method or metric	Who uses it	Why include it
Best single judge	Kohli 2026, preprint	Matched or beat a nine-judge panel worth about two independent votes
Majority vote at equal compute	Wang et al. 2023; Choi et al. 2025	Choi et al. prove discussion alone does not raise expected correctness
Rater model with independent errors	White paper; Dawid and Skene 1979 (unverified)	Shows what counting shared mistakes once adds (P1)
Panel from different model families (PoLL)	Verga et al. 2024, preprint	Beat one large judge at over 7 times lower cost
Discussion; standard debate with a fixed judge	Du et al. 2024; Khan et al. 2024	What the project replaces; isolates impartial choice and the harness

Method or metric	Who uses it	Why include it
Effective number of judges (independent votes a panel is worth); error correlation; CAPA (chance-adjusted shared mistakes)	Hossain et al. 2026, preprint; Goel et al. 2025, venue not confirmed; Kim et al. 2025, ICML	Shared error per condition (P1, P2); held-out model families can still share errors
Calibration error (gap between confidence and accuracy); accuracy on the disputed step	Guo et al. 2017; ProcessBench	The debate threshold (eq. 3) needs calibrated posteriors; P4 concerns single steps
Wrong decisions that get through, usefulness (rescaled accuracy) and false flags at 0.5%, 2% and 5% audit budgets	Greenblatt et al. 2024; Schaeffer et al. 2026	Main red-team result; prior work used 0.5% and 2%

Fixes to the white paper

- **Testbed B.** Reconcile MEQA's question count: 2,243 in the paper, 2,093 in the README. Suggested: "multi-hop event questions from our MEQA pipeline, extended to several documents with outdated, forged or selectively chosen reports, including maritime scenarios seeded with public vessel tracks." No formal export-control review was done.
- **Testbed A.** "Agents of known skill" needs a mechanism, since imitating a weaker model had only "mediocre success" (van der Weij et al. 2024). Suggested: "agents whose skill we measure on held-out items; k of N agents receive the same flawed hint (Chen et al., 2025)."
- **Red team.** Training colluders is ACL-2 or higher, beyond Greenblatt et al. 2024, whose red team prompted and scaffolded GPT-4. Suggested: "a red team at four levels, from prompting to training arbitrary attack strategies (Korbak et al., 2025)."
- **Audit rate.** Griffin et al. 2024 (preprint) cast control evaluations as partially observable stochastic games (players see only part of the state) and reduce them to zero-sum games, so existing algorithms can find protocols with the best safety-usefulness trade-offs. Suggested, as the mathematical reasoning the program officer requested: "We set the audit rate and the share of planted debates by solving a game between the red team and our protocol (Griffin et al., 2024)."
- **Baselines.** Replace "averaging with equal weights" with the draft paragraph's list.
- **Testbed C.** Suggested: "replication outcomes released in 2026 serve as human audits, treated as noisy labels with an estimated error rate," matching the draft's P6.
- **Novelty.** The search found no study that tracks shared error as one hint reaches more models, and no control evaluation that protects a judge's decision in a debate. It was not exhaustive, and its 2025–2026 sweep was under-sampled because the search budget ran out, so both gaps are provisional. Suggested: "To our knowledge, no prior control evaluation protects decisions made by a judge in a debate."
- **Section 2, line 39.** The Amayuelas et al. 2024 (preprint) abstract gives no number for "sharply cut"; cite one from the full paper or soften it.

Draft evaluation paragraph

> All data are public or synthetic. (A) Agents whose skill we measure on held-out items solve math and code problems. One flawed hint goes to k of N agents (Chen et al., 2025), so we know who shares a mistake, and expert labels mark the first wrong step (Zheng et al., 2024). (B) We extend our MEQA pipeline (Li et al., 2024) to several documents with outdated, forged or selectively chosen reports. Maritime scenarios start from public vessel tracks, but our generator fixes the true answer. (C) For scientific claims and peer review, replication outcomes serve as human audits, and papers with planted errors (Schroter et al., 2008) as automatic audits. Following AI control (Greenblatt et al., 2024), a red team at four levels (Korbak et al., 2025) makes debaters and judges collude, recognize planted debates, and attack only when they expect no audit. We count wrong decisions that get through at audit budgets of 0.5% to 5%. Baselines are the best single judge, majority vote at equal compute, a rater model with independent errors, a panel from different model families (Verga et al., 2024), discussion (Du et al., 2024), and standard debate (Khan et al., 2024).